EMBEDDED SYSTEMS CAPSTONE PROJECT

WOMEN SAFETY WRISTBAND SYSTEM

GPS-Based Location Sharing and GSM Emergency SMS Alert

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Abstract

This project presents a wearable Women Safety Wristband System designed to support emergency communication through a physical SOS button, GPS-based location acquisition and GSM-based short message service (SMS). The prototype uses an ESP32 microcontroller, a NEO-6M GPS receiver, a SIM900A GSM module, an OLED display and a buzzer. When the wearer presses the SOS button, the controller reads available GPS coordinates, prepares a map location link and sends an emergency SMS to a preconfigured contact. The OLED and buzzer provide local feedback. The project focuses on a simple, single-action emergency-alert workflow. The supplied project presentation reports successful functional demonstration, including receipt of the location link by an emergency contact. Numerical measurements for message latency, GPS accuracy, reliability, battery life and cost were not provided; these are therefore identified as evaluation items rather than achieved results.

Keywords: Women safety, ESP32, NEO-6M, SIM900A, GPS, GSM, SOS alert, SMS, location sharing.

1. Introduction

1.1 Background

In an emergency, unlocking a phone, opening an application and manually sharing a location can take time. A wearable device with a dedicated SOS button can simplify this process. This project combines a physical emergency trigger with GPS location acquisition and GSM messaging so that a trusted contact can receive an alert and a map link.

1.2 Problem Statement

The project addresses the difficulty of quickly communicating distress and location through a conventional phone-based workflow. It aims to provide a compact prototype in which one deliberate button press initiates local feedback and an SMS-based location alert.

1.3 Aim

To design and demonstrate a wearable embedded safety prototype that detects an SOS button press, obtains available GPS coordinates, shares a location link through GSM SMS and provides status feedback through an OLED display and buzzer.

1.4 Objectives

- Integrate the ESP32, SOS pushbutton, NEO-6M GPS receiver, SIM900A GSM module, OLED display and buzzer.
- Implement an emergency workflow initiated by a physical SOS button.
- Generate a location-sharing message from available GPS coordinates.
- Provide local visual and audible feedback during the alert process.
- Demonstrate the workflow by checking device feedback and SMS receipt on a target phone.

1.5 Scope and Limitations

The implemented scope described in the project presentation is manual SOS activation, GPS-based location sharing, GSM SMS transmission, OLED status feedback and buzzer feedback. Automatic risk prediction, machine learning, location-history analysis and geofencing are not described as completed features and should be presented as future enhancements unless they have been implemented and tested separately. GPS operation may be affected by poor satellite visibility, and SMS delivery depends on SIM configuration, cellular coverage and stable power.

2. Hardware and Software Requirements

Component	Purpose
ESP32 microcontroller	Controls the SOS workflow and coordinates the connected modules.
NEO-6M GPS receiver	Provides latitude and longitude when a valid satellite fix is available.
SIM900A GSM module	Sends the emergency SMS through a cellular network.
SOS pushbutton	Allows the wearer to initiate an emergency alert.
OLED display	Displays system state or alert status.
Buzzer	Provides local audible feedback when an SOS event is triggered.
Power supply and wiring	Powers the controller and modules; the GSM supply must support transmission current peaks.

2.1 Software Requirements

Include the actual development environment and libraries used (for example, Arduino IDE and the libraries used for GPS parsing or the OLED).

3. System Design and Methodology

3.1 System Architecture

The ESP32 acts as the central controller. The SOS pushbutton provides the input event, the NEO-6M supplies position data, the SIM900A handles SMS communication, and the OLED and buzzer provide local feedback.

Block / Module	Connection or data flow
SOS pushbutton → ESP32	Button press triggers the emergency routine.
NEO-6M GPS → ESP32	GPS data is read and checked for a valid position.
ESP32 → SIM900A	The controller sends modem commands and the emergency SMS text.
ESP32 → OLED	Status information is shown to the wearer.
ESP32 → Buzzer	Local audible indication is activated for the SOS event.
SIM900A → Emergency contact	SMS contains the location link when valid coordinates are available.

Insert the team's actual circuit/block diagram here. Check every pin number against the physical wiring and firmware; do not include an unverified pin map.

3.2 Working Procedure

- Power on the device and initialize the ESP32, GPS interface, GSM serial interface, OLED, button input and buzzer.
- Keep the device in its ready state while monitoring the SOS button and incoming GPS data.
- When the SOS button is pressed, validate the button event to reduce accidental duplicate triggers.
- Read the latest GPS data and check whether the coordinates represent a valid fix.
- Construct an SMS containing the emergency message and the map link based on the available coordinates.
- Submit the message through the SIM900A to the configured emergency contact.
- Activate local feedback and update the OLED with an accurate status message.

3.3 Algorithm

- Start.
- Initialize modules and enter ready state.
- Has the SOS button been pressed? If no, continue monitoring.
- Read GPS data and check fix validity.
- Prepare the emergency SMS and location link when coordinates are available.
- Send the SMS through the GSM module and activate the local alert.
- Display the appropriate status and return to monitoring.

4. Implementation and Results

4.1 Implemented Functions

Function	Evidence described in the project presentation	Interpretation
SOS trigger	The physical SOS button initiates the workflow.	Manual emergency trigger demonstrated.
GPS location	The workflow obtains coordinates and builds a map link.	Location-sharing function demonstrated; numerical accuracy not reported.
GSM SMS	Emergency contacts received the location link during the demonstration.	Successful receipt was reported for the demonstration; total attempts are not stated.
OLED	The display shows “Msg sent to emergency contact.”	Local status feedback demonstrated; confirm that firmware reports the true modem status.

Buzzer	The buzzer provides audible feedback during an emergency.	Local audible indication included in the prototype.

4.2 Prototype Evidence

- complete hardware prototype
- ESP32 and connected modules
- OLED status during an alert
- the received SMS with its map link.

5. Discussion and Limitations

The prototype brings together a physical SOS trigger, GPS location acquisition, GSM messaging and local feedback. Its main usability advantage is that the wearer can initiate an alert with one deliberate action. However, the system depends on a valid GPS fix and usable cellular service. GPS reception can be poor indoors or in obstructed locations, while SMS delivery can fail because of weak coverage, SIM configuration or power-supply instability. An OLED message alone does not prove that the recipient received the SMS. The firmware should distinguish between an alert request, modem acceptance and recipient delivery confirmation where confirmation is available.

This is a prototype, not a certified safety product. It should supplement rather than replace emergency services and personal safety planning.

6. Future Enhancements

These items are proposed enhancements and should not be described as implemented features unless the team has built and tested them.

- Location-history logging with timestamps, with appropriate privacy and access controls.
- Risk-area or isolated-area warnings based on verified geographic data.
- Machine-learning-based route-risk analysis, only after collecting suitable data and validating a model.
- Improved handling of invalid GPS data, missing cellular service and failed SMS submission.
- More compact wearable packaging and improved battery-life evaluation.

7. Conclusion

The Women Safety Wristband System demonstrates a basic wearable emergency-alert workflow using an ESP32, NEO-6M GPS receiver and SIM900A GSM module. A physical SOS button initiates location sharing through SMS, while the OLED and buzzer provide local feedback. The project presentation reports successful functional demonstration, including receipt of a location link by an emergency contact. Further testing is needed to quantify response time, GPS accuracy, message reliability and battery performance. Location history, geofencing and machine-learning-based risk analysis are suitable future enhancements.

8.References

IoT-Based Wrist Band for Women Safety

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9. What We Learned from our Capstone Project

Through this project, we learned how to integrate different hardware components, such as the microcontroller, GPS, GSM, SOS button, OLED display, and buzzer, into one system. We gained practical experience in programming, circuit connections, and troubleshooting. We also understood the importance of a proper GPS signal, cellular connectivity, and reliable power supply for sending emergency alerts. This project taught us that safety devices need careful testing, accuracy, and reliability. Overall, it improved our embedded systems knowledge and gave us ideas for future improvements to the wristband.